

GARISSA, Kenya

WMO: 637230

Lat: 0.464S

Lon: 39.648E

Elev: 148

StdP: 99.56

Time Zone: 3.00 (E03)

Period: 94-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.5	22.1	12.7	9.3	32.1	13.8	10.0	31.9	14.9	28.7	13.5	29.2	1.9	180	0.452

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	11.0	37.5	23.7	36.9	23.7	36.2	23.7	26.2	32.3	25.7	31.8	25.4	31.5	3.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.9	20.3	27.8	24.4	19.7	27.5	24.1	19.3	27.3	82.1	32.3	80.3	32.1	78.9	31.8	34.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.0	10.0	8.7	DB	15.5	39.0	3.7	1.1	12.8	39.8	10.6	40.5	8.5	41.1	5.8	41.9	(4)
(5)			WB	14.9	27.8	2.7	0.8	12.9	28.4	11.4	28.9	9.9	29.4	7.9	30.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	29.1	29.6	30.4	31.1	30.5	29.2	27.9	27.2	27.4	28.2	29.3	29.0	29.0
	DBStd	1.57	0.91	0.93	0.93	1.15	1.10	0.96	0.92	0.88	0.90	1.15	1.25	1.09
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6957	608	571	654	616	596	537	533	538	547	599	570	589
	CDD18.3	3916	349	337	396	366	338	287	274	280	297	341	320	331
	CDH23.3	46093	4411	4519	5427	4713	4056	2989	2587	2698	3191	4038	3614	3849
	CDH26.7	22781	2313	2524	3075	2452	1943	1284	1064	1158	1527	2011	1635	1794
Wind	WSAvg	3.7	2.0	2.6	3.0	3.8	4.5	5.2	5.2	5.2	4.9	4.0	2.2	1.8
Precipitation	PrecAvg	357	16	2	39	68	16	5	4	8	5	32	106	55
	PrecMax	924	247	15	172	176	75	32	18	75	51	168	350	174
	PrecMin	130	0	0	0	5	0	0	0	0	0	0	14	2
	PrecStd	173	48	4	41	44	20	7	5	16	10	44	75	41
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.2	38.2	38.3	37.6	36.1	34.2	33.6	34.0	35.2	36.2	36.1	36.1
		MCWB	22.7	23.2	24.0	24.5	24.2	22.3	21.6	21.5	22.4	23.4	24.2	23.1
	2%	DB	36.3	37.2	37.7	36.8	35.2	33.5	32.9	33.1	34.3	35.6	35.1	35.1
		MCWB	22.9	23.3	23.9	24.4	24.0	22.3	21.6	21.3	22.0	23.3	24.0	23.4
	5%	DB	35.7	36.6	37.0	36.0	34.4	32.9	32.2	32.4	33.7	34.9	34.3	34.3
		MCWB	23.2	23.3	24.0	24.3	23.8	22.3	21.4	21.2	21.8	23.2	24.1	23.7
	10%	DB	34.8	35.8	36.2	35.1	33.7	32.1	31.5	31.8	32.9	34.1	33.3	33.4
		MCWB	23.4	23.3	24.0	24.4	23.6	22.1	21.3	21.2	21.8	23.0	24.2	23.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.0	26.1	26.5	26.8	25.9	24.6	23.2	23.7	24.2	25.4	26.2	26.4
		MCDB	33.1	34.3	34.2	32.5	33.3	30.5	30.7	29.6	31.6	32.0	31.7	31.3
	2%	WB	25.2	25.1	25.7	26.1	25.2	23.7	22.7	22.7	23.4	24.7	25.6	25.7
		MCDB	32.6	33.8	32.7	31.9	31.9	30.2	30.0	29.4	30.8	31.3	31.1	30.8
	5%	WB	24.7	24.5	25.3	25.7	24.7	23.3	22.3	22.2	22.9	24.3	25.2	25.3
		MCDB	32.1	33.3	32.2	31.4	31.1	29.7	29.2	28.9	30.0	30.6	30.5	30.4
	10%	WB	24.4	24.1	25.0	25.3	24.3	22.9	22.0	21.8	22.4	24.0	24.9	25.0
		MCDB	31.6	32.9	31.8	30.8	30.7	29.2	28.5	28.3	29.5	30.2	30.1	30.1
Mean Daily Temperature Range	5% DB	MDBR	11.1	11.7	11.0	9.9	9.6	9.4	9.5	9.7	10.4	10.3	9.0	9.4
		MCDBR	12.3	12.7	11.7	10.7	10.4	10.1	10.3	10.3	11.1	11.0	10.2	10.6
		MCWBR	3.1	2.9	2.3	2.1	2.2	2.3	2.4	2.1	2.3	2.3	2.1	2.5
	5% WB	MCDBR	10.5	11.5	10.6	9.6	9.5	9.1	9.4	9.5	10.3	9.9	8.8	8.9
		MCWBR	3.0	3.0	2.5	2.5	2.5	2.7	2.6	2.5	2.6	2.6	2.5	2.6
Clear-Sky Solar Irradiance	taub	0.463	0.479	0.466	0.417	0.409	0.471	0.469	0.430	0.409	0.410	0.437	0.465	
	taud	2.211	2.152	2.218	2.396	2.419	2.182	2.176	2.301	2.350	2.391	2.336	2.233	
	Ebn at Noon	870	861	865	887	870	802	809	864	901	911	888	863	
	Edh at Noon	150	161	150	122	115	142	145	132	129	125	131	145	
All-Sky Solar Radiation	RadAvg	5.88	6.20	6.27	5.95	5.62	5.13	5.17	5.46	5.92	6.09	5.67	5.63	
	RadStd	0.29	0.28	0.20	0.20	0.17	0.21	0.18	0.17	0.22	0.19	0.29	0.24	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page